

* Higher Order Derivatives:

The derivative f' of a function f itself a function and hence may have a derivative of its own. If f' is differentiable, then its derivative is denoted by f'' and is called the second derivative of f . As long as we have differentiability, we can continue the process of differentiating to obtain third, fourth, fifth, and even higher derivatives of f . These are called higher derivatives or successive derivatives.

$$f'(x) = 12x^3 - 6x^2 + 2x - 4$$

$$f''(x) = 36x^2 - 12x + 2$$

$$f'''(x) = 72x - 12$$

$$f^{(4)}(x) = 72$$

$$f^{(n)}(x) = 0 \quad (n \geq 5).$$

The Chain Rule:

If g is differentiable at x and f is differentiable at $g(x)$, then the composition $f \circ g$ is differentiable at x .

$$\therefore \frac{d}{dx} (f \circ g)(x) = \frac{d}{dx} [f(g(x))] = f'(g(x)) g'(x).$$

Example: Evaluate $\frac{dy}{dx}$ of $y = \tan^2 x$.

Solution: Given, $y = \tan^2 x$ $\begin{cases} f(x) = x^2 \\ g(x) = \tan x \end{cases}$

$$\Rightarrow y = [\tan x]^2$$

Applying Chain Rule:

$$\frac{d}{dx} [f(g(x))] = f'(g(x)) g'(x)$$

$$\frac{d}{dx} [\tan^2 x] = 2 \tan x \cdot \sec^2 x.$$

* Logarithmic Differentiation:

$$\frac{d}{dx} [\ln x] = \frac{1}{x}$$

$$\frac{d}{dx} [\ln(x^2 + 1)] = \frac{2x}{x^2 + 1}$$

Example: Evaluate the derivative of:

$$y = \frac{x^3 \sqrt[3]{7x-14}}{(1+x^2)^4}$$

$$\Rightarrow \ln(y) = \ln \left[\frac{x^3 \cdot \sqrt[3]{7x-14}}{(1+x^2)^4} \right]$$

$$\Rightarrow \ln(y) = \ln[x^2 \cdot \sqrt[3]{7x-14}] - \ln(1+x^2)^4$$

$$\Rightarrow \ln(y) = \ln(x^2) + \ln(\sqrt[3]{7x-14}) - 4\ln(1+x^2)$$

$$\Rightarrow \ln(y) = 2\ln(x) + \frac{1}{3}\ln(7x-14) - 4\ln(1+x^2)$$

$$\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = \frac{2}{x} + \frac{7}{3(7x-14)} - \frac{8x}{1+x^2}$$

$$\Rightarrow \frac{dy}{dx} = y \left[\frac{2}{x} + \frac{7}{3(7x-14)} - \frac{8x}{1+x^2} \right]$$

$$\therefore \frac{dy}{dx} = \frac{x^2 \sqrt[3]{7x-14}}{(1+x^2)^4} \left[\frac{2}{x} + \frac{7}{3(7x-14)} - \frac{8x}{1+x^2} \right] \quad (A)$$

Explicit Function: An equation of the form $y=f(x)$ is said to define explicitly as a function of x because the variable y appears alone on one side of the equation and does not appear on the other side at all.

$$y = \frac{x-1}{x+1}$$

Implicit Function: In general, it is not necessary to solve an equation for y in terms of x in order to differentiate the functions defined implicitly by the equation.

$$5y^2 + \sin(y) = x^2$$

* Implicit Differentiation:

$$5y^2 + \sin(y) = x^2$$

$$\Rightarrow \frac{d}{dx}(5y^2 + \sin y) = \frac{d}{dx}(x^2)$$

$$\Rightarrow 10y \frac{dy}{dx} + \cos y \frac{dy}{dx} = 2x$$

$$\Rightarrow \frac{dy}{dx}(10y + \cos y) = 2x$$

$$\therefore \frac{dy}{dx} = \frac{2x}{10y + \cos y}$$

* H.W.: (1) $\cos(xy) = e^{x+y}$.

(2) $xy + e^y = 0$.

* Leibnitz Theorem: Leibnitz Rule is a generalization of the product rule. It states that if the function $f(x)$ and $g(x)$ are differentiable n times, then their product $f(x) \cdot g(x)$ is also differentiable n times.

$$\begin{aligned} \frac{d^n}{dx^n}(fg) &= \frac{d^n f}{dx^n} g + nC_1 \cdot \frac{d^{(n-1)} f}{dx^{(n-1)}} \cdot \frac{dg}{dx} + nC_2 \cdot \frac{d^{(n-2)} f}{dx^{(n-2)}} \cdot \frac{d^2 g}{dx^2} \\ &+ \dots + nC_k \cdot \frac{d^{(n-k)} f}{dx^{(n-k)}} \cdot \frac{d^k g}{dx^k} + \dots + f \cdot \frac{d^n g}{dx^n} \\ &= \sum_{k=0}^n nC_k \cdot \frac{d^{(n-k)} f}{dx^{(n-k)}} \cdot \frac{d^k g}{dx^k} \end{aligned}$$

Example: Evaluate $\frac{d^n}{dx^n}(x^2 e^{ax})$ using Leibnitz Theorem.

Solution: Let, $f(x) = e^{ax}$; $g(x) = x^2$.

$$\begin{aligned}\frac{d^n}{dx^n}(x^2 e^{ax}) &= \frac{d^n}{dx^n}(e^{ax}) \cdot x^2 + nC_1 \cdot \frac{d^{(n-1)}}{dx^{(n-1)}}(e^{ax}) \cdot \frac{d}{dx}(x^2) \\ &\quad + nC_2 \cdot \frac{d^{(n-2)}}{dx^{(n-2)}}(e^{ax}) \cdot \frac{d^2}{dx^2}(x^2) + 0. \\ &= a^n e^{ax} \cdot x^2 + n \cdot a^{(n-1)} \cdot e^{ax} \cdot 2x + \frac{n(n-1)}{2} \cdot a^{n-2} \cdot 2e^{ax} \\ &= a^n e^{ax} \cdot x^2 + 2xn \cdot a^{(n-1)} \cdot e^{ax} + n(n-1) \cdot a^{n-2} \cdot e^{ax} \quad \underline{\text{(Ans)}}$$

H.W.: If $y = x^{n-1} \cdot e^{1/x}$ prove that:

$$y_n = (-1)^n \cdot \frac{e^{1/x}}{x^{n+1}}$$

Practice Sheet:

Exercise 2.4 $\rightarrow (5-20), 21, 24$.

Exercise 2.6 $\rightarrow$ Example $\rightarrow 5, 6$,

Ex-3, 4, 7-40, 43-54.

Exercise 2.3 $\rightarrow 29, 30, 31, 32, 41-48$.

Example: If $y = \tan^{-1}(x)$ then show that:
 $(1+x^2)y_{n+2} + 2(n+1)x y_{n+1} + n(n+1)y_n = 0$.

Solution: Given, $y = \tan^{-1}(x)$

$$\Rightarrow y_1 = \frac{1}{1+x^2}$$

$$\Rightarrow (1+x^2)y_1 = 1$$

Again Differentiating with respect to x :

$$\frac{d}{dx} [(1+x^2)y_1] = \frac{d}{dx} [1]$$

$$\Rightarrow (1+x^2)y_2 + y_1 \cdot \frac{d}{dx}(1+x^2) = 0$$

$$\Rightarrow (1+x^2)y_2 + 2xy_1 = 0 \text{ ————— (1)}$$

Applying Leibnitz Theorem in equation (1),

$$\frac{d^n}{dx^n} [(1+x^2)y_2] + \frac{d^n}{dx^n} [2xy_1] = 0$$

$$\Rightarrow \frac{d^n}{dx^n} (y_2) \cdot (1+x^2) + n \cdot \frac{d^{n-1}}{dx^{n-1}} (y_2) \cdot \frac{d}{dx}(1+x^2)$$

$$+ n \cdot \frac{d^{n-2}}{dx^{n-2}} (y_2) \cdot \frac{d^2}{dx^2}(1+x^2) + \frac{d^n}{dx^n} (y_1) \cdot 2x$$

$$+ n \cdot \frac{d^{n-1}}{dx^{n-1}} (y_1) \cdot \frac{d}{dx}(2x) = 0$$

$$\Rightarrow (1+x^2)y_{n+2} + n(2x)y_{n+1} + \frac{n(n-1)}{2} \cdot y_n \cdot 2 +$$

$$y_{n+1} \cdot 2x + n \cdot y_n \cdot 2 = 0$$

$$\Rightarrow (1+x^2)y_{n+2} + 2x(n+1)y_{n+1} + (n^2 - n + 2n)y_n = 0$$

$$\Rightarrow (1+x^2)y_{n+2} + 2x(n+1)y_{n+1} + n(n+1)y_n = 0 \text{ [Shown]}$$

* Parametric Equation: Parametric equation is a type of equation that employs an independent variable called parameter and in which dependent variables are defined as continuous functions of the parameter.

Parametric Differentiation:

Example: The equation of ellipse is specified by: $x = a \cos t$, $y = b \sin t$.

The slope/gradient of the ellipse is:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{b \cos t}{-a \sin t} = -\frac{b}{a} \cot(t).$$

$$\therefore \text{at } t = \frac{\pi}{6}: \left. \frac{dy}{dx} \right|_{t=\frac{\pi}{6}} = -\frac{b}{a} \cot\left(\frac{\pi}{6}\right) = -\frac{\sqrt{3}b}{a}.$$

* A tangent line to a parameter curve will be horizontal at those points where $\frac{dy}{dt} = 0$ and $\frac{dx}{dt} \neq 0$ i.e. $\frac{dy}{dx} = 0$ at such points.

* A tangent line to a parametric curve will be vertical at those points where $\frac{dy}{dt} \neq 0$ and $\frac{dx}{dt} = 0$ i.e. there exists infinite slope.

* When $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$, then we get intermediate form of the slope. We call such point as a singular point.

Example: In a distance flight, an experimental paper airplane follows the tangency of the particle: $x = t - 3 \sin t$; $y = 4 - 3 \cos t$. But crashes into a wall at time, $t = 10$ s.

(a) At what times, ^{was} the plane flying horizontally?

(b) At what times, ^{was} the plane flying vertically?

Solution: The tangency of the particle is given as: $x = t - \sin 3t$
 $y = 4 - 3 \cos t$.

$$\text{Here, } \frac{dy}{dt} = 3 \sin t$$

$$\frac{dx}{dt} = 1 - 3 \cos 3t$$

Since, the plane was flying horizontally:

$$\frac{dy}{dt} = 3 \sin t = 0$$

$$\Rightarrow \sin t = 0$$

$$\Rightarrow \sin t = \sin(0 + n\pi)$$

$$\therefore t = n\pi.$$

There are four solutions on the interval $0 \leq t \leq 10$. Therefore, $t = 0, \pi, 2\pi, 3\pi$

Here, $\frac{dx}{dt} \neq 0$ at points $t = 0, \pi, 2\pi, 3\pi$.

(b) The airplane was flying vertically at those times when: $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} \neq 0$. Setting

$$\frac{dx}{dt} = 0 \Rightarrow 1 - 3\cos t = 0 \Rightarrow \cos t = \frac{1}{3}$$

The equation has three solutions in the time interval $0 \leq t \leq 10$.

$$t = \cos^{-1}\left(\frac{1}{3}\right), 2\pi - \cos^{-1}\left(\frac{1}{3}\right), 2\pi + \cos^{-1}\left(\frac{1}{3}\right)$$

Here, $\frac{dy}{dt} = 3\sin t$ is not zero at these points. (Ans)

Local Linear Approximation:

A function that is differentiable at x_0 is sometimes said to be locally linear at x_0 .

The line that best approximates the graph of f in the vicinity of $P(x_0, f(x_0))$ is the tangent line to the graph of f at x_0 , given by the equation: $y \approx f(x_0) + f'(x_0)(x - x_0)$

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

This is called the local linear approximation of f at x_0 .

Problem: Find the local linear approximation of $f(x) = \sqrt{x}$ at $x_0 = 1$.

Solution: Given that, $f(x) = \sqrt{x}$
 $f(x_0) = \sqrt{1} = 1$.

$$\text{Again, } f'(x) = \frac{1}{2\sqrt{x}}$$

$$f'(x_0) = \frac{1}{2}$$

Thus, the local linear approximation at $x_0 = 1$ is:

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

$$\Rightarrow \sqrt{x} \approx 1 + \frac{1}{2}(x - 1) \quad (\text{Ans})$$

Problem: H.W.: Find the local linear approximation of $f(x) = \sin x$ at $x_0 = 0$.

Example: Find the gradient at the point $(1, 2)$ on the curve whose equation is given by:
 $x^3 - 5xy^2 + y^3 + 1 = 0$.

Solution: Given: $x^3 - 5xy^2 + y^3 + 1 = 0$

$$\Rightarrow \frac{d}{dx}(x^3 - 5xy^2 + y^3 + 1) = 0$$

$$\Rightarrow 3x^2 - 5y^2 - 10xy \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$$

$$\Rightarrow (-10xy + 3y^2) \frac{dy}{dx} = -3x^2 + 5y^2$$

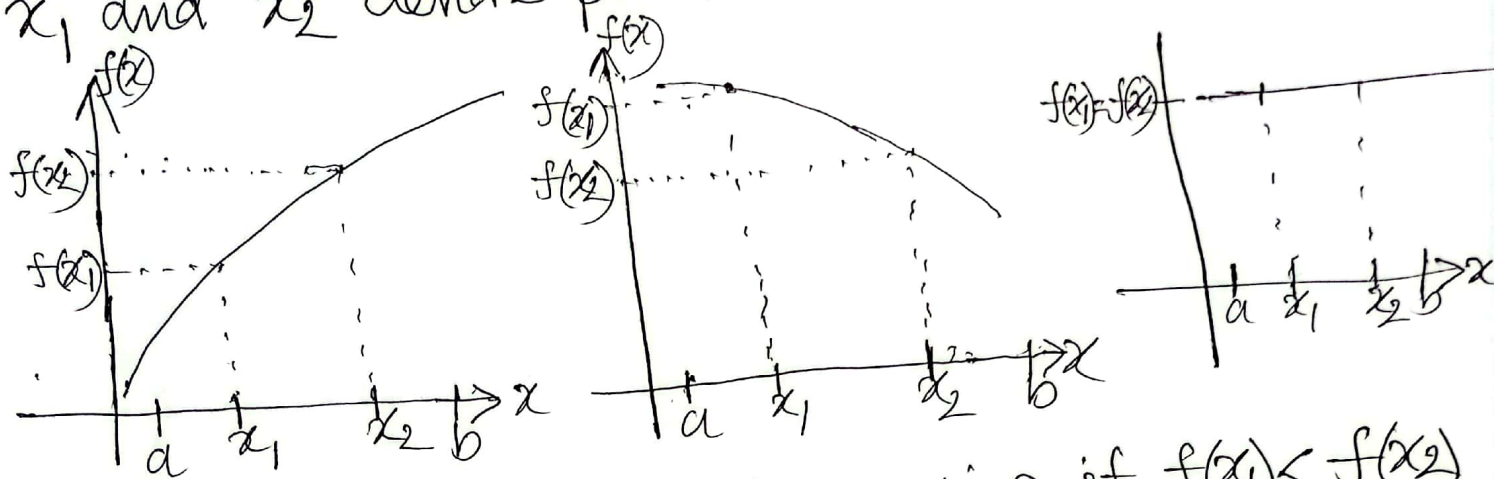
$$\Rightarrow \frac{dy}{dx} = \frac{5y^2 - 3x^2}{3y^2 - 10xy}$$

At Point $(1, 2)$:

$$\left. \frac{dy}{dx} \right|_{(1,2)} = \frac{5(2)^2 - 3(1)^2}{3(2)^2 - 10 \cdot 1 \cdot 2} = \frac{20 - 3}{12 - 20} = \frac{-17}{8}$$

* Increasing and Decreasing Functions:

Let, f be defined on an interval and let x_1 and x_2 denote points on that interval:



- (a) The function $f(x)$ is increasing if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$.
- (b) The function $f(x)$ is decreasing if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$.
- (c) The function $f(x)$ is constant if $f(x_1) = f(x_2)$ whenever $x_1 < x_2$.

* Theorem: Let, f be a function that is continuous on a closed interval on the open interval (a, b) .

- (a) If $f'(x) > 0$ for every value of x in (a, b) , then f is increasing on $[a, b]$.
- (b) If $f'(x) < 0$ for every value of x in (a, b) , then f is decreasing on $[a, b]$.
- (c) If $f'(x) = 0$ for every value of x in (a, b) , then f is constant on $[a, b]$.

Example: Find the intervals on which $y = x^3$ is increasing or decreasing.

Solution: Given:

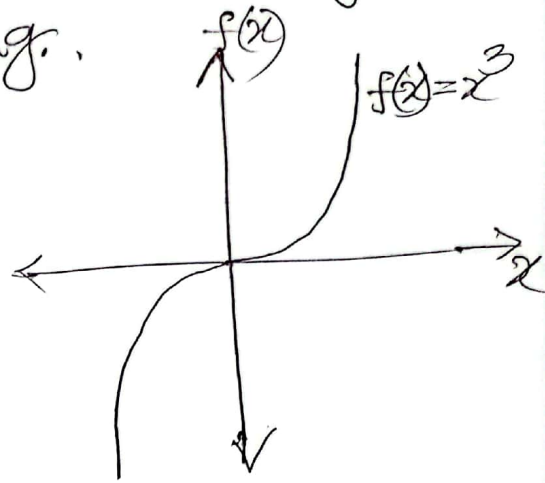
$$f(x) = x^3$$

$$f'(x) = 3x^2$$

Thus: $f'(x) > 0$ if $x < 0$

$f'(x) > 0$ if $x > 0$.

Since, f is continuous everywhere, f is increasing on $(-\infty, 0]$ and f is increasing on $(-\infty, +\infty)$.



Example: Find the intervals on which $f(x) = 3x^4 + 4x^3 - 12x^2 + 2$ is increasing or decreasing.

Solution: Given: $f(x) = 3x^4 + 4x^3 - 12x^2 + 2$.

$$f'(x) = 12x^3 + 12x^2 - 24x$$

$$= 12x(x^2 + x - 2)$$

$$= 12x(x^2 + 2x - x - 2)$$

$$= 12x(x+2)(x-1)$$

$$x = 0, -2, 1$$

The sign analysis of f' is given below:

Interval	$12x(x+2)(x-1)$	$f'(x)$	Conclusion
$x < -2$	$(-)(-)(-)$	$-$	Decreasing on $(-\infty, -2)$
$-2 < x < 0$	$(-)(+)(-)$	$+$	Increasing on $(-2, 0)$
$0 < x < 1$	$(+)(+)(-)$	$-$	Decreasing on $(0, 1)$
$x > 1$	$(+)(+)(+)$	$+$	Increasing on $(1, +\infty)$

$\therefore f$ is increasing on the intervals

$\therefore f$ is increasing on the intervals $(-2, 0)$ and $(1, +\infty)$ and decreasing on the intervals $(-\infty, -2)$ and $(0, 1)$.

Practice Problems:

Exercise 3.5: $5 \rightarrow 10, 11 \rightarrow 16$

Exercise 10.1: $53, 54, 45 \rightarrow 52$

Exercise 4.1: $15 \rightarrow 32 (a, b)$

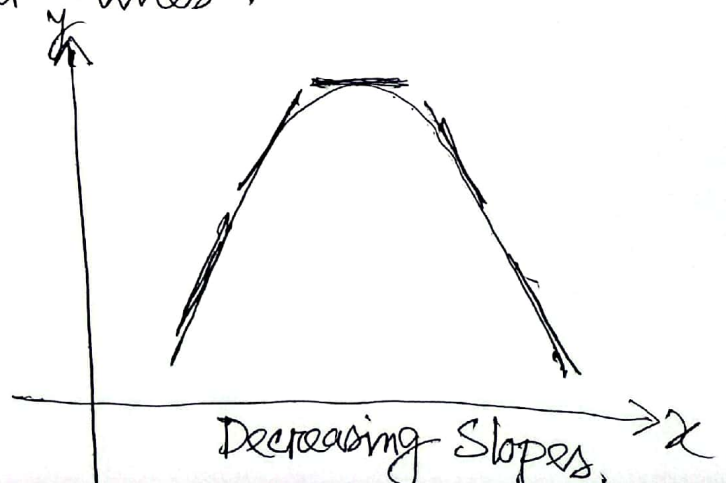
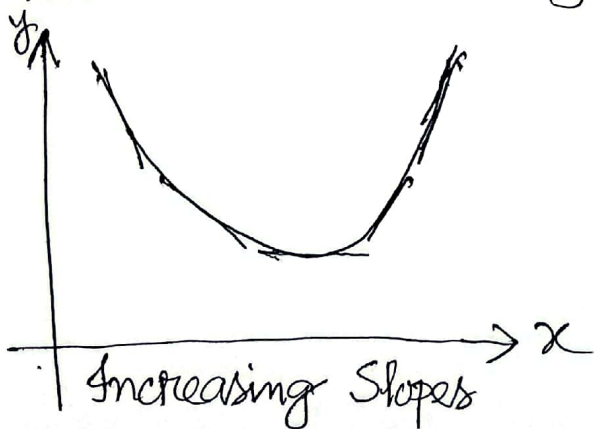
Exercise 4.2: $7-14, 25-32, 33-50$.

* Concavity: Although the sign of the derivative of f reveals where the graph of f is increasing or decreasing, it does not reveal the direction of curvature. On intervals where the graph of f has upward curvature we say that f is concave up and on intervals where the graph of f has downward curvature we say that f is concave down.

The following two points suggest two ways to characterize the concavity of a differentiable function on an open interval:

① f is concave up on an open interval if its tangent lines have increasing slopes on that interval and is concave down if they have decreasing slopes.

② f is concave up on an open interval if its graph lies above its tangent lines on that interval and is concave down if it lies below its tangent lines.



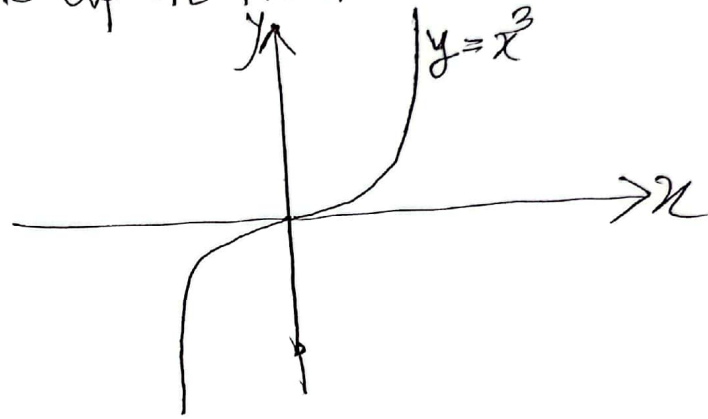
* Theorem: Let f be twice differentiable on an open interval.

(a) If $f''(x) > 0$ for every value of x in the open interval, the f is concave up on that interval.

Example: Let, $f(x) = x^3$

$$f'(x) = 3x^2$$

$$f''(x) = 6x$$



for $x < 0$; $f''(x) < 0$ on the open interval. Again,

for $x > 0$; $f''(x) > 0$ on the open interval.

$\therefore f(x) = x^3$ is concave down on the interval $(-\infty, 0)$ and concave up on the interval $(0, +\infty)$.

* Inflection point: If f is continuous on an open interval containing a value of x_0 and if f changes the direction of its concavity at the point $(x_0, f(x_0))$, then we say that f has an inflection point at x_0 and we call the point $(x_0, f(x_0))$ on the graph of f an inflection point of f .

Example: If the function $f(x)$ is given as $f(x) = x^3 - 3x^2 + 1$; then find the intervals on which

- f is increasing.
- f is decreasing.
- f is concave up.
- f is concave down.
- Find all inflection points.

Solution: Given that:

$$f(x) = x^3 - 3x^2 + 1$$

$$f'(x) = 3x^2 - 6x = 3x(x-2)$$

$$f''(x) = 6x - 6 = 6(x-1)$$

The sign analysis of these derivatives are shown in the following table:

Interval	$3x(x-2)$	$f'(x)$	Conclusion
$x < 0$	$(-)(-)$	$+$	f is increasing on $(-\infty, 0)$
$0 < x < 2$	$(+)(-)$	$-$	f is decreasing on $(0, 2)$
$x > 2$	$(+)(+)$	$+$	f is increasing on $(2, +\infty)$

The inflection points are found by setting:

$$\begin{aligned} f''(x) &= 0 \\ \Rightarrow 6(x-1) &= 0 \\ \therefore x &= 1 \end{aligned}$$

Interval	$6(x-1)$	$f''(x)$	Conclusion
$x < 1$	$-$	$-$	Concave Down $(-\infty, 1)$

Example: $f(x) = \frac{1}{20}x^5 - \frac{1}{3}x^4 + \frac{1}{6}x^3 + 3x^2 + 5x - 2$

$$f'(x) = \frac{1}{4}x^4 - \frac{4}{3}x^3 + \frac{1}{2}x^2 + 6x + 5$$

$$f''(x) = x^3 - 4x^2 + x + 6$$

$$= (x+1)(x-2)(x-3)$$

The inflection points are found by setting

$$f''(x) = 0$$

$$\Rightarrow x^3 - 4x^2 + x + 6 = 0$$

$$\Rightarrow x^3 + x^2 - 5x^2 - 5x + 6x + 6 = 0$$

$$\Rightarrow x^2(x+1) - 5x(x+1) + 6(x+1) = 0$$

$$\Rightarrow (x+1)(x^2 - 5x + 6) = 0$$

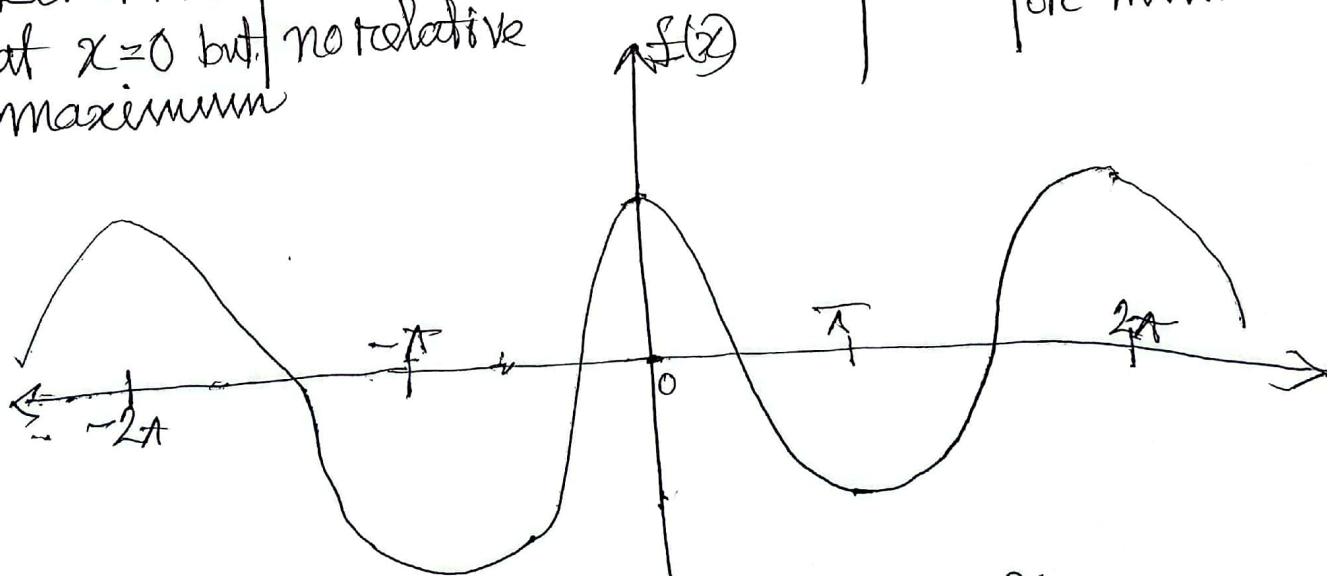
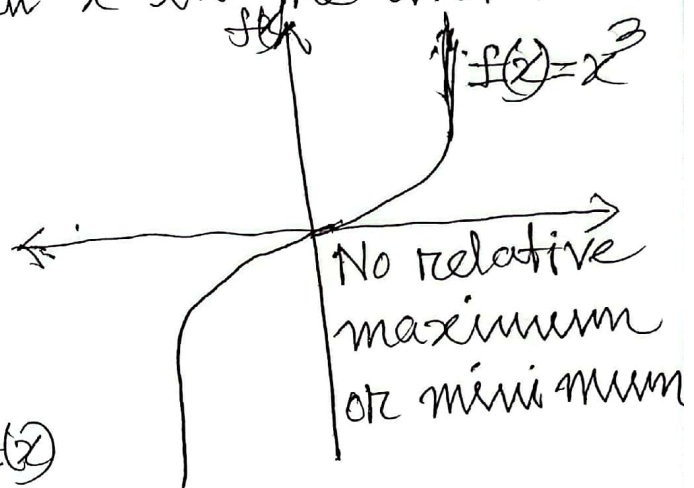
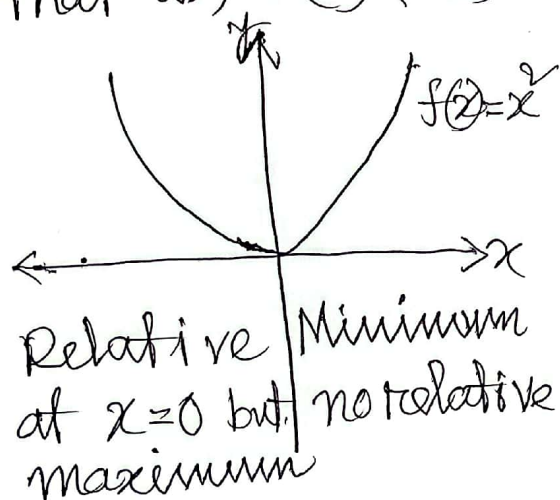
$$\Rightarrow (x+1)(x-2)(x-3) = 0$$

$$\therefore x = -1, 2, 3.$$

Intervals	$(x+1)(x-2)(x-3)$	sign of f''	Conclusion
$x < -1$	$(-)(-)(-)$	$-$	Concave down $(-\infty, -1)$
$-1 < x < 2$	$(+)(-)(-)$	$+$	Concave up $(-1, 2)$
$2 < x < 3$	$(+)(+)(-)$	$-$	Concave Down $(2, 3)$
$x > 3$	$(+)(+)(+)$	$+$	Concave up $(3, +\infty)$

*Relative Maxima and Minima:

A function f is said to have a relative maximum at x_0 if there is an open interval containing x_0 on which $f(x_0)$ is the largest value, that is $f(x_0) \geq f(x)$ for all x in the interval. Similarly, f is said to have a relative minimum at x_0 if there is an open interval containing x_0 on which $f(x_0)$ is the smallest value, that is, $f(x_0) \leq f(x)$ for all x in the interval.



Relative maxima at all even multiples of π and relative minima at all odd multiples of π .

Critical Point: In general, we define critical point for a function f to be a point in the domain of f at which either the graph of f has a horizontal tangent line or f is not differentiable. To distinguish between the two types of critical points we call x ~~stationary~~ stationary point of f if $f'(x)=0$.

Example: Find all critical points of $f(x) = x^3 - 3x + 1$.

Solution: The function f being a polynomial, is differentiable everywhere, so its critical points are all stationary points. To find these points, we must solve the equation $f'(x)=0$. Since.

$$\begin{aligned} f'(x) &= 3x^2 - 3 = 0 \\ &\Rightarrow 3(x^2 - 1) = 0 \\ &\therefore x = -1, 1. \end{aligned}$$

$\therefore$ the critical points occur at $x = -1, x = 1$

Example: Find all critical points of

$$f(x) = 3x^{5/3} - 15x^{2/3}.$$

Solution: Given, $f(x) = 3x^{5/3} - 15x^{2/3}$.

$$f'(x) = 5 \cdot x^{2/3} - 10x^{-1/3} = x^{-1/3}(5x - 10) \\ = \frac{5(x-2)}{x^{1/3}}$$

We see from this that $f'(x) = 0$ if $x = 2$ and $f'(x)$ is undefined at $x = 0$. Thus $x = 0$ and $x = 2$ are critical points and $x = 2$ is a stationary point.

First Derivative Test: Let's suppose that f is continuous at a critical point x_0 .

(a) If $f'(x) > 0$ on an open interval extending left from x_0 and $f'(x) < 0$ on an open interval extending right from x_0 , then f has a relative maximum at x_0 .

(b) If $f'(x) < 0$ on an open interval extending left from x_0 and $f'(x) > 0$ on an open interval extending from x_0 , then f has a relative minimum at x_0 .

© If $f'(x)$ has the same sign on an open interval extending left from x_0 as it does on an open interval extending right from x_0 , then f does not have a relative maximum or minimum at x_0 .

Example: The function $f(x) = 3x^{5/3} - 15x^{2/3}$ has a relative maximum at $x=0$ and a relative minimum at $x=2$. Confirm this using the first derivative test.

Solution: Given: $f(x) = 3x^{5/3} - 15x^{2/3}$

$$f'(x) = \frac{5(x-2)}{x^{1/3}}$$

The sign of f' changes from $+$ to $-$ at $x=0$ so there is a relative maximum at this point. The sign changes from $-$ to $+$ at $x=2$, there is a relative minimum at that point.

"Second Derivative Test": Let's suppose that f is twice differentiable at the point x_0 :

(a) If $f(x_0) = 0$ and $f''(x_0) > 0$, then f has a relative minimum at x_0 .

⑥ If $f(x_0) = 0$ and $f''(x_0) < 0$, then f has a relative maximum at x_0 .

⑦ If $f'(x_0) = 0$ and $f''(x_0) = 0$ then the test is inconclusive, i.e. f may have a relative maximum, a relative minimum, or neither at x_0 .

Example: Find the relative extrema of:

$$f(x) = 3x^5 - 5x^3.$$

Solution: Given: $f(x) = 3x^5 - 5x^3$

$$\therefore f'(x) = 15x^4 - 15x^2$$

$$\therefore f''(x) = 60x^3 - 30x = 30x(x^2 - 1)$$

Solving: $f'(x) = 0$

$$\Rightarrow x = 0, -1, 1 \rightarrow \text{stationary points.}$$

	$\frac{30x(2x^2-1)}{-30}$	$\frac{f''(x)}{-}$	$\frac{\text{Second derivative test}}{f \text{ has a relative maximum}}$
$x = -1$			

$x = 0$	0	0	Inconclusive
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$x = 1$	30	+	f has relative minimum.
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The test is inconclusive at $x = 0$, so we will try the first derivative test at that point.

Interval	$\frac{15x^2(x+1)(x-1)}{(+)(+)(-)}$	$\frac{f'(x)}{-}$
$-1 < x < 0$		
$0 < x < 1$		

Since there is no sign change in f' at $x=0$, there is neither a relative maximum nor a relative minimum at that point.

General Guideline for sketching graph of a function:

- 1) Domain
- 2) Intercepts
- 3) Symmetry Test.
- 4) Asymptotes.
- 5) Intervals of Increase or decrease.
- 6) Local Maximum and Minimum.
- 7) Concavity and Points of inflection.
- 8) Sketch the curve.